Plan for today:

1. Menti on understanding from last time (5-10 min)
2. Neumann BC code and effect of dx (5-10 min)
3. Lecture (BCs) (20 min)
4. Break 10 min
5. Exercises

Join menti 3882 6599

Boundary conditions

Frederik Dalby

Modelling Fall 2025

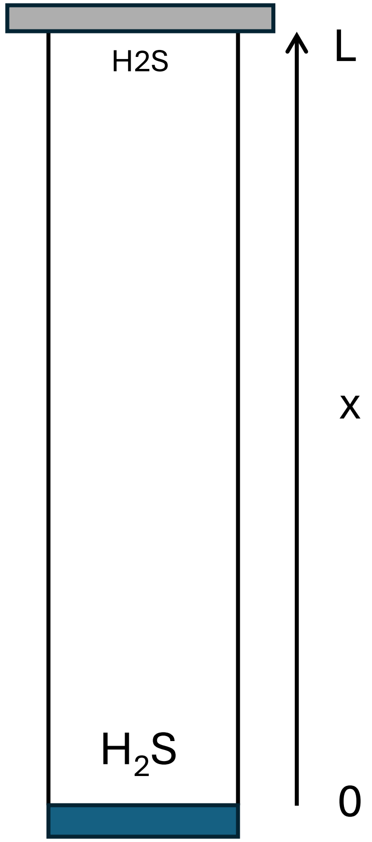
Three overall types

Boundary Condition	Mathematical Form	Physical Meaning	
Dirichlet	$C = \text{given value}$	Fixed concentration at boundary	(first type)
Neumann	$\frac{\partial C}{\partial x} = \text{given value}$	Fixed flux (including zero flux)	(second type)
Robin	$aC + b\frac{\partial C}{\partial x} = \text{given value}$	Mixed condition; flux proportional to concentration difference	(third type)

Use constitutive equations at boundary (newtons law, mass transfer coefficient, reaction kinetics)

We can have different type of boundary conditions in each end: Example with Dirichlet + Neumann BC

A well with contaminated water at the bottom is emitting H_2S. There is always 1000 ppm H_2S at the bottom. On the top of the well a concrete lid ensures that H_2S cannot escape.



Neumann BC (fixed flux)

$$J(t, x = L) = -D\frac{\partial C_{(t,x=L)}}{\partial x} = 0$$

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

Governing eq

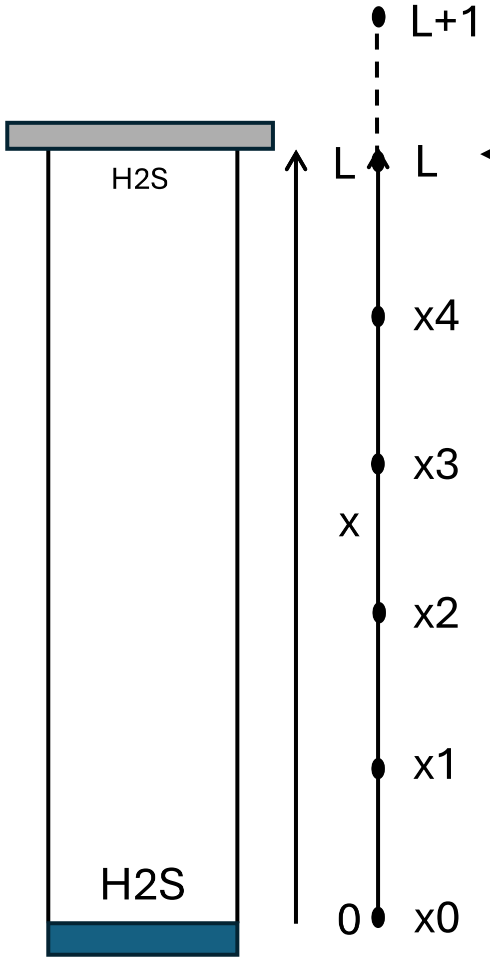
2 BC (for all time points)

1 IC (for every ODE)

Dirichlet BC

$$C(t, x = 0) = 1000$$

ghost point



Neumann BC (fixed flux)

$$J = -D\,\frac{\partial C_{(t,z=L)}}{\partial x} = 0$$

$$\frac{\partial C}{\partial x} = 0$$

$$\frac{c_{L+1} - c_{L-1}}{2\Delta x} = 0$$

$$c_{L+1} = c_{L-1}$$

$$\frac{\partial C}{\partial t} = D\,\frac{\partial^2 C}{\partial x^2}$$

Governing eq

Dirichlet BC

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$ Governing eq

$\downarrow$ reformulate to discretized form

$$\frac{\partial C}{\partial t} = D\frac{c_{i-1} - 2c_i + c_{i+1}}{\Delta x^2}$$

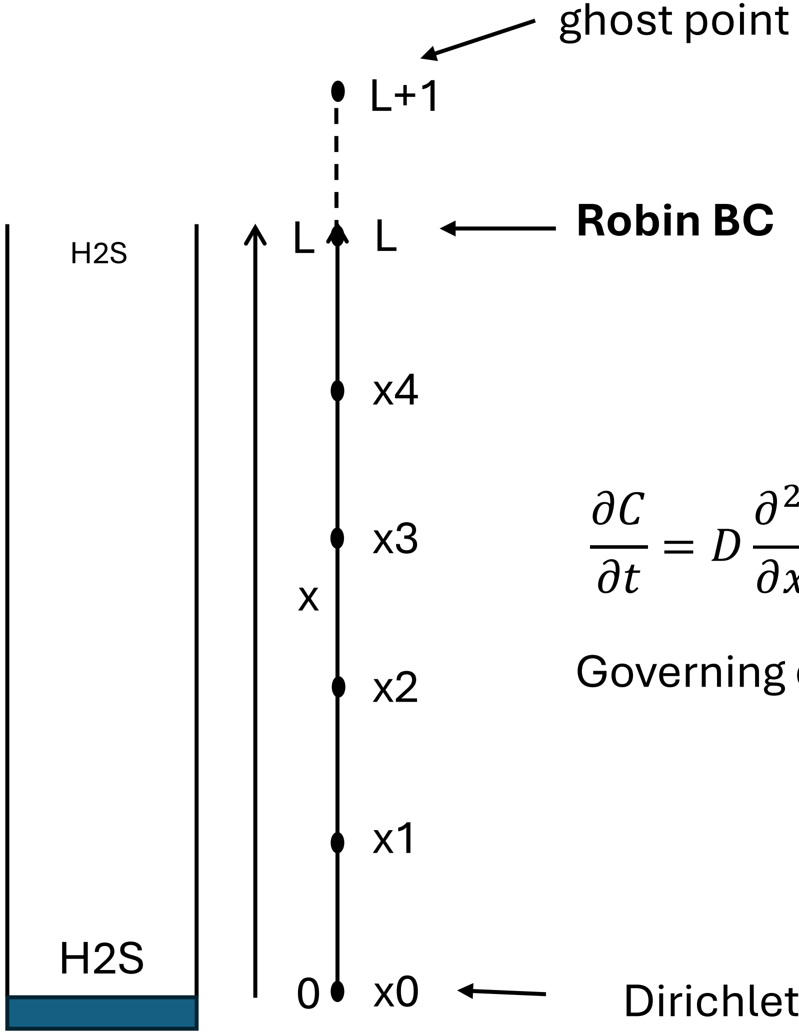
$\downarrow$ At the top of the well

$$\frac{\partial C(t, x = L)}{\partial t} = D\frac{c_{L-1} - 2c_L + c_{L+1}}{\Delta x^2}$$

$\downarrow$ Substitute $c_{L+1} = c_{L-1}$

$$\frac{\partial C(t, x = L)}{\partial t} = D\frac{c_{L-1} - 2c_L + c_{L-1}}{\Delta x^2} = D\frac{2(c_{L-1} - c_L)}{\Delta x^2}$$

A well with contaminated water at the bottom is emitting H_2S. There is always 1000 ppm H_2S at the bottom. The top of the well is open to the air with a mild wind over the well. H_2S is transferred to the bulk air phase by convection



$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

Governing eq

$$J = -D\frac{dC_{(t,x=L)}}{dx} = f(C_{(t,x=L)}, C_{bulk})$$

$$-D\frac{dC(t, x = L)}{dx} = k_c \cdot (C_{(t,x=L)} - C_{bulk})$$

$$-D\frac{c_{L+1} - c_{L-1}}{2\Delta x} = k_c \cdot (C_L - C_{bulk})$$

$$c_{L+1} - c_{L-1} = \frac{2\Delta x k_c \cdot (C_L - C_{bulk})}{-D}$$

$$c_{L+1} = -\frac{2\Delta x k_c \cdot (C_L - C_{bulk})}{D} + c_{L-1}$$

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$ Governing eq

$\downarrow$ reformulate to discretized form

$$\frac{\partial C}{\partial t} = D\frac{c_{i-1} - 2c_i + c_{i+1}}{\Delta x^2}$$

$\downarrow$ At the top of the well

$$\frac{\partial C(t, x = L)}{\partial t} = D\frac{c_{L-1} - 2c_L + c_{L+1}}{\Delta x^2}$$

Substitute $$c_{L+1} = -\frac{2\Delta x k_c \cdot (C_L - C_{bulk})}{D} + c_{L-1}$$

$$\frac{\partial C(t, x = L)}{\partial t} = D\frac{c_{L-1} - 2c_L + (-\frac{2\Delta x k_c \cdot (C_L - C_{bulk})}{D} + c_{L-1})}{\Delta x^2} = 2D\frac{c_{L-1} - c_L}{\Delta x^2} - 2k_c\frac{c_L - c_{bulk}}{\Delta x}$$

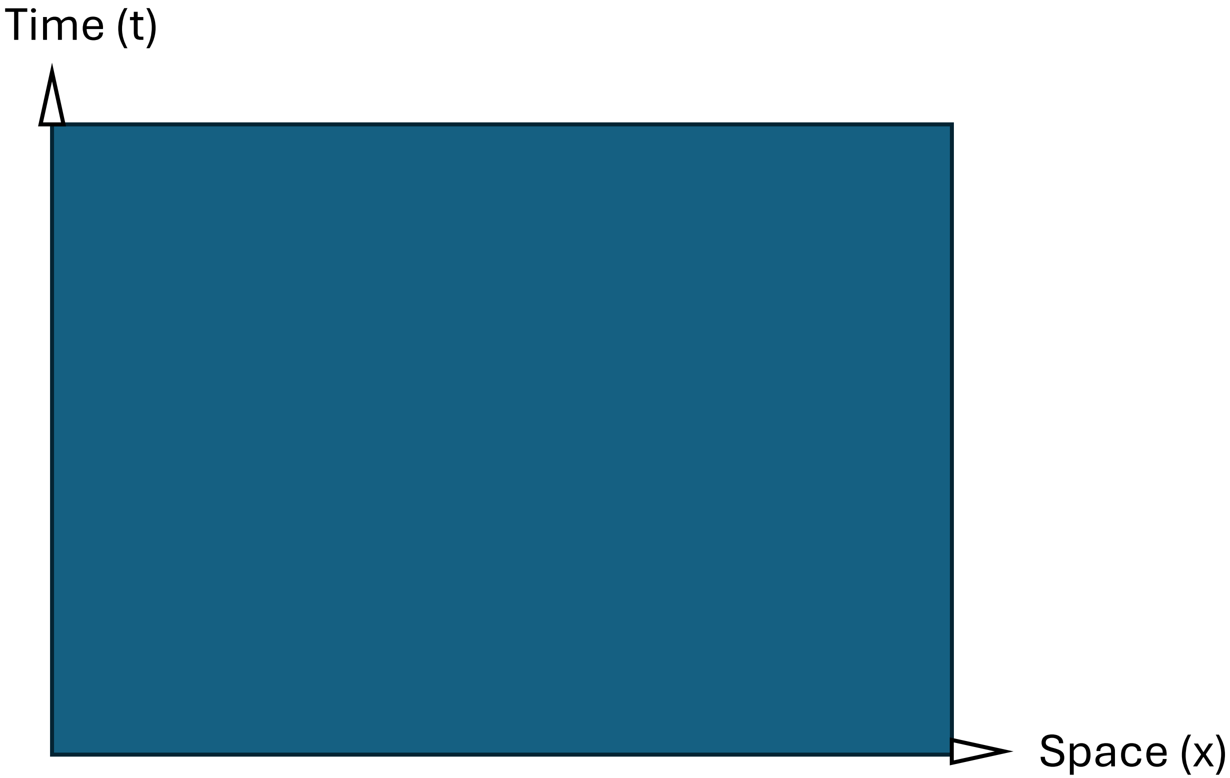
ill-posed problems

1. **A solution does not exists (contradicting boundary or intial conditions)**
2. Non-unique solution
3. Unstable solution

Governing EQ

$$\frac{\partial C}{\partial t} = D\,\frac{\partial^2 C}{\partial x^2}$$

C(t, x = 0) = 5



C(t=0, x) = 10

Governing EQ

$$\frac{\partial C}{\partial t} = D\,\frac{\partial^2 C}{\partial x^2}$$

$$\frac{\partial C}{\partial t} = -v\frac{\partial C}{\partial x}$$

Algorithms for solving text problems

1. Select PDE coordinate system
2. Eliminate uneccesary terms to get the governing equation
3. Reformulate to discretized form (for method of lines)
4. Figure out what the BC’s and IC’s are
5. Implement in python by handling interior and BC nodes separately